SAFETY DATA SHEET

1. IDENTIFICATION OF THE MATERIAL/MIXTURE AND THE COMPANY/ENTERPRISE

1.1 Product identification
TSM Ceramic INSULATION COATING

1.2 The proper identified usage or contradicted usage of material or mixture:
An organic binder for inner and outer industrial usage. Application area: architecture, thermal engineering, forwarding.

1.3 Information of the data sheet's transporter
Thermográf Ltd.
Budapest 1122, St. Városmajor 43/C. basement 2.
Tel: +36-92-898-005

Ukkeramzit Ltd.
Ukraine, Uzhgorod 88000
Pankievicha St. 108/A
1.3.1. Liable person: *György Csilik*
E-mail: csilik.gyorgy@tsmceramic.com

1.4 Hot line:
Health Toxicological Information Service (HTIS)
Budapest 1096, Square Nagyvárad 2.
Tel.: +36 1 476 6464, +36 80 201 199

2. DEFINITION OF DANGER

2.1. Classification of the material of mixture
Classification by REACH order:

Not classified as a dangerous mixture!

R sentences that refer to the danger/risk of material: -

S sentences that refer to safe usage of material:
S 2 – Not for children!
S 26 – If it contacted with eyes, immediately wash it away with water

2.2. Labeling elements
R sentences that refer to the danger/risk of material: -

S sentences that refer to safe usage of material:
S 2 – Not for children!
S 26 – If it contacted with eyes, immediately wash it away with water

2.3. Other dangers
Don't store it in frosty places. If used by proper usage, the material doesn't have any harmful conditions.

3. COMPOSITION OR INFORMATION ABOUT COMPOSITION

3.1. Mixture:

Description: flexible insulation material with micro sized hollowed and rutile ceramic balls filled with acrylic polymer and rubber binder

4. FIRST AID MEASURES

4.1. Description of first aid measures

4.1.1. SWALLOWING:

What to do:

4.1.1.1. Wash out the mouth of the injured person and give him large amount of water

4.1.1.2. Call a doctor and show them the label or the safety data sheet

4.1.1.3. One shall not feed anything with the injured person through their mouth

4.1.2. INHALATION:

What to do:

4.1.2.1. Take the injured to fresh air, loosen up their clothes and place them in a comfortable position

4.1.3. SKIN CONTACT:

What to do:

4.1.3.1. Take down the polluted clothes

4.1.3.1. Wash the skin with plenty of soaped water (for 15 minutes!)

4.1.4. EYE CONTACT:

What to do:

4.1.4.1. Wash the eye with water by stretching the corners of the eyelid and moving the eyeball at same time (for about 10 minutes!)

4.1.4.2. If symptoms are shown, immediately go to a doctor

4.2. The most important (acute and delayed) symptoms and effects:

If used by proper usage, the material doesn't have any harmful conditions.

4.3. Indication of the required immediate medical care or special treatment:

No data available

5. FIRE SAFETY MEASURES

Not combustible!

MSZ-EN 15824:2009 Flammability class: C

5.1. Extinguishing agent:

5.1.1. Proper extinguishing agent:

Foam, dry extinguishing powder, carbon-dioxide, water spray

5.1.2. Unsuitable extinguishing agent: -

5.2. Special dangers of the material or mixture:

During fire accidents toxic gases may be freed which, when inhaled, can be harmful

5.3. Recommendation for firemen:

As stated in regulations, full protective clothing may be worn with breath apparatus that is not connected with the air. The product itself is inflammable but they should be cooled with water spray when set on fire

6. MEASURES WITH RANDOM EXPOSITION

6.1. Personal precautionary measures, personal protective equipments and emergency regulations:

6.1.1. Not emergency personnel:

Only a person that is fully aware of what to do, trained and equipped with proper protective equipment may stay on the place of accident

6.1.2. Emergency personnel:

Wear protective equipments that are stated in regulations, try to assure effective ventilation

6.2. Ecological precautionary measures:

The material that got into the environment should be treated according to proper regulations. The material and its refugees should not be let into waters, the ground or sewers. If the pollution happened, the competent authorities should be immediately notified.

6.3. Territorial delimitation and methods and materials of disposing:

The material that was freed to the environment should be absorbed with absorbing material (e.g.: universal binder) and then until it will get utilized or disposed should be stored in a labeled container. During the collecting, storing and disposing use of personal protective equipment is recommended.

6.4. Reference to other paragraphs:

For further information see the 8th and the 13th paragraphs.

7. HANDLING AND STORING

7.1. Precautionary measures for safe handling:

Compliance of the usual hygiene rules is obligatory

Technical measures:

Don't let out in closed space and don't spray it

Provide effective ventilation and if necessary local suction

Fire and explosion hazards regulation:

No special regulations

7.2. Conditions of safe storing with possible contrarieties:

Conditions of safe storing:

The product should be stored only in its original, closed and labeled container. The storage room should be ventilated and cleared properly!

Don't let it to get into the ground

Should be stored in cool, dry place, away from cold and high temperature

Recommended storage temperature: 1-26ºC

If the temperature is higher than 60ºC, its quality may be damaged

Contrariety materials: N/A

Type of the material used for containing/storing: plastic bucket; should be stored in its original, container.

7.3. Specified end use (uses):

No special instruction available

8. CONTROL OF EXPOSITION/PERSONAL PROTECTION

8.1. Parameters of control:

DNEL		Way of exposition	Frequency of exposition	Comments
Employee	User	Dermal	Short (acute) Long (repeated)	
		Inhalation	Short (acute) Long (repeated)	
		Oral	Short (acute) Long (repeated)	

PNEC			Frequency of exposition	Comments
Water	Air	Ground		
			Short (onefold) Long (continuous)	
			Short (onefold) Long (continuous)	
			Short (onefold) Long (continuous)	

8.2. Inspection of exposition:

According to the 7th (6) paragraph of the 25/2000 (IX. 30.) EUM-SZCSM joint decree the employer should minimize the extent of exposition to the lowest level of the technical standards with hazardous materials which are not regulated by hazard limits.

8.2.1. Proper technical inspection

During work special attention should be given to avoid the pouring of the mixture onto the floor, clothing, skin and the eye.

Take care of proper ventilation.

Skin and eye contact should be avoided.

During work breaks and after work thorough hand washing is recommended.

Usage of skin protecting cream is recommended.

8.2.2. Personal protection:

8.2.2.1. Eye/face protection: usage of goggles that meets the regulations is recommended.

8.2.2.2. Skin protection:

8.2.2.2.1. Hand protection: usage of butyl rubber gloves that meets the regulations is recommended.

8.2.2.2.2. Other: usage of protecting clothing that meets the regulations is recommended.

8.2.2.3. Protection of respiratory tracts: not necessary.

8.2.2.4. Heat hazards: not known.

8.2.3. Environmental exposition controls:

No special instructions

The regulations described in the 8th paragraph within usual circumstances are apply to workmanlike activities and proper usage.

If the work is done in different circumstances or in exceptional circumstances then from all the other things to do and from personal protecting equipment the experts should be asked.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1.1. Information on the basic physical and chemical properties:

Parameter	Testing method	Comments
9.1.2. External features:	white liquid	
9.1.3. Smell:	light, characteristic	
9.1.4. Odor threshold:	-	
9.1.5. pH value:	-	
9.1.6. Melting/freezing point	-	
9.1.7. Boiling range:	-	
9.1.8. Fire point:	-	
9.1.9. Evaporation rate:	-	
9.1.10. Fire hazardousness:	Inflammable	
9.1.11. Weight of product:	-	
9.1.12. Vapor density:	-	
9.1.13. Density:	MSZEN 101506:1999	
9.1.14: Solubility:	dilutable with distilled water	
9.1.15. Partition coefficient: n-octanol/water:		
9.1.16. Auto ignition temperature:	Inflammable	
9.1.17. Decomposition point:	Does not decompose	
9.1.18. Viscosity:	-	
9.1.19. Explosion hazardous properties:	-	
9.1.20. Oxidation properties:	-	

9.2. Other information:

The usage of the material is recommended at a temperature from -35ºC to +260ºC.

10. STABILITY AND REACTIVITY

10.1. Reactivity:

Not known.

10.2. Chemical stability:

At normal temperature and in average work conditions the material is stabile.

10.3. Possibilities of dangerous reactions:

No data available.

10.4. Avoided conditions:

Not known.

10.5. Incompatible materials:

Not known.

10.6. Hazardous decomposition products:

Carbon-monoxide, carbon-dioxide

11. TOXICOLOGICAL DATA

11.1. Information on toxicological effects

Acute toxicity: not known.
Irritation: not known.
Caustic effect: not known.
Sensitization: not known.
Repeated dose toxicity: not known.
Carcinogenic effect: not known.
Mutagenicity: not known.

Properties that impair reproduction: not known.

11.2. Summary of results of clinical examinations:
No data available.
11.3. Relative toxicological information:
No data available.
11.4. Information on possible ways of exposition:
Swallowing, inhaling, skin and eye contact.
11.5. Symptoms of physical, chemical or toxicological characteristics:
No data available.
11.6. Delayed, immediate and chronic effects of short and long-term exposition:
If used by proper usage, the material doesn't have any harmful conditions.
11.7. Effects from interaction:
No data available.
11.8. Lack of individual data:
No information.
11.8.1. Other information:
No data available.

12. ECOLOGICAL INFORMATION

12.1. Toxicity
No data available. Don't let the product into the environment without control.
12.2. Persistence and degradability:
No data available.
12.3. Bioaccumulation abilities:
No data available.
12.4. Mobility in the ground:
No data available.
12.5. Results of the PBT and vPvB testing:
No data available.
12.6. Other harmful effects:
No data available.

13. ASPECTS ON DISPOSITION

13.1. Methods of waste managing:
The product should be treated and disposed as told in the 98/2001 (VI.15.) Government regulation [modified by: 340/2004 (XII. 22.) Gov. regulation; 313/2005 (XII. 25.) Gov. regulation] and in the 16/2001 (VII. 18.) Gov. regulation [modified by: 22/2004(XII. 11.) Gov. regulation].
13.2. Information on the material's disposition:
For information about recycling ask the manufacturer. Waste list code: 08 02 03 water suspension with ceramic materials.
13.3. Information on disposition of the packing:
The cleaned packing is recyclable. The unclear packing can be disposed just like the product.
13.4. Physical or chemical aspects that affect the waste management:
N/A
13.5. Instructions on wastewater treatment:
N/A
13.6. Possible special precautions for waste management:
No data available.

14. INFORMATION ON TRANSPORTATION:

The product is not a dangerous consignment!

14.1. UN-number:

-

14.2. Proper shipping name according to UN:

-

14.3. Hazard classification(s):

-

14.4. Package group:

-

14.5. Environmental dangers:

-

14.6. Special precautionary measures for the user:

-

14.7. Bulk delivery on the 73/78 annex of the MARPOL and the IBC codex:

-

15. INFORMATION ON REGULATION

15.1. Safety, public health and ecological regulations/laws on the product:

15.1.1. International REACH regulation:

The 1907/2006/EK decrees of the European Parliament and the Council (18th of December, 2006) on registration, evaluation, authorization and restriction of chemical substances (REACH), on foundation of the European Chemicals Agency, modifying the 1999/45/EK directive and also repealing the 793/93/EGK council regulation, the 1488/92/EK safety regulation, the 76/769/EGK council directive, the 91/155/EGK, the 93/67/EGK, the 93/105/EK and the 2000/21/EK safety directive.

15.1.2. International CLP regulation:

THE 1272/2008/EK DECREE OF THE EUROPEAN PARLIAMENT AND THE COUNCIL (16th of December, 2008) on classification of materials and mixtures, on labeling and wrapping, on modifying and repealing the 67/548/EGK and the 1999/45/EK directive and on modifying the 1907/2006/EK regulation.

15.1.3. THE 453/2010/EU DECREE OF THE COUNCIL (20th of May, 2010) on modifying the 1907/2006/EK decree on registration, evaluation, authorization and restriction of chemical substances (REACH).

15.1.4. Decrees on dangerous substances:

The year 2000 law on chemical safety [modified by: the year 2004 340/2004 law; the year 2004 CXL. law; the year 2005 CXXVII. law] and its concerning laws: the 44/2000 (XII. 27) decree of MoH [modified by 33/2004 (IV. 26) EszCsM; 60/2005 (XII. 20) decree of MoH; 3/2006 (I. 26) decree of MoH; the 1/2005 (I. 7) decree of MARD; the 61/2004 (VIII. 11) decree of EsCsM; the 73/2004 (VIII. 11 decree of EszCsM; the 26/2007 (VI. 7) decree of EUM]

15.1.5. Regulations on hazardous waste:

The 98/2001 (VI. 15.) Gov. decree [modified by: the 340/2004 (XII. 22) Gov. decree; the 313/2005 (XII. 25) Gov. decree], the 16/2001 (VII. 18) MoE decree in the 16/2001 (VII. 18 MoE decree) [modified by: 22/2004 (XII. 11) MoEW decree]

15.1.6. Decrees on water pollution:

The 220/2004 (VII. 21) Gov. decree (modified by: 368/2004 (XII. 26) Gov. decree; 340/2004 (XII. 22) Gov. decree; 208/2006 (X. 16) Gov. decree)

15.1.7. Regulations on occupational safety:

The year 1993 XCIII decree on occupational safety, its modifications and its concerning NM and Mi.iM decrees.

15.1.8. Regulations on chemical safety of workplaces:

The 25/2000 (IX. 30) Ei.iM-SaFA joint decree.

15.2. Chemical safety rating:

Was not made.

16. OTHER INFORMATION

Data on revision of the safety data sheet: -
Full texts of the abbreviations in the safety data sheet:
DNEL: Derived no effect level. PNEC: Predicted no effect concentration. CMR effects: carcinogenicity, mutagenicity and reproductive toxicity. PBT: persistent, bioaccumulative and toxic. n.s: not specified. n.a: not applicable.
Literature used/references: -
Recommended user limitations (the not obligatory proposal of the supplier): -
This safety data sheet was made by documentations given by the manufacturer and it fits to the year 2000 XXV law on chemical safety and its modification, also its concerning decrees, the 44/2000 MoH decree and the modification of the regulation of the 33/2004 (V. 26) ESZCSM decree (on certain proceedings and rules on detailed activities with hazardous materials and mixtures).
The information, data and recommendations on this safety data sheet, which on the date of publishing was considered accurate, correct and workmanlike, were made by honest works of competent experts. They can be provided as only a guide to the usage of material, without lack of fullness. While used in certain circumstances, other considerations which were not mentioned here may be necessary. In view of this the author of the safety data sheet, because not knowing the circumstances of using and treating the product, is not responsible directly or indirectly for the material's quality and doesn't give a guarantee on that all the information, data and recommendation on the safety data sheet will be fully accurate at the time of using the product. The author of the safety data sheet and its publisher are not responsible for anything that is written here and for any damage, loss, injury, accident as well as any similar or related happening that can be related to the usage of information given here. The

deliberation of the validity of the information given here and the ascertainment on the concrete usage and treatment of the material is the responsibility of the performer of the activities. The user is obligated to follow all those legislative requirements which refer to the usage of the product.